

class auditing → next week
→ next thursday

Routing Protocol Classification → later

#Flooding

Forwarding system Routing path পাঠ্য শিষ্ট।
Forwarding vs Routing

Distance Vector Algorithm

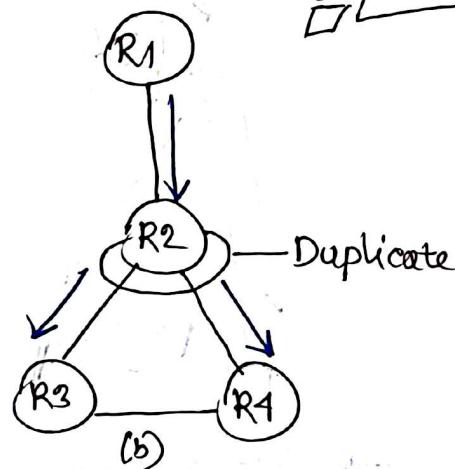
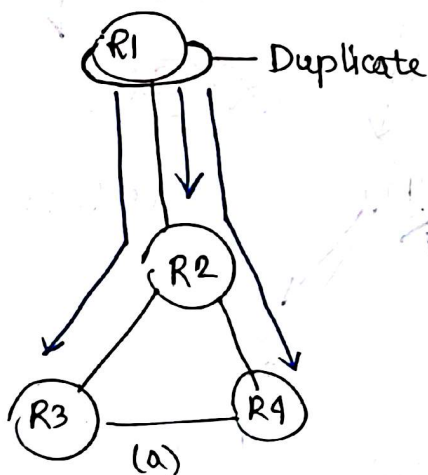
Wireless Network

└ Wireless Nodes / Sensors

└ Aggregated Network

Flooding {good algorithm}
└ has own set of issues

❗ Flooding has many mechanisms.



4.43: Source duplication (a) vs
In-network duplication (b)

Lab Assignment
(7 তারিখ)

FTP/HTTP
~~Server~~
Server

20 Marks

Time 1 week

Computer Exam
AND/OR
WRITTEN Exam

13 April or
20 April

☒ Broadcast Storm (!!)

↳ Sequence-number-controlled-flooding {from book}

☒ Reverse Path Forwarding (RPF)

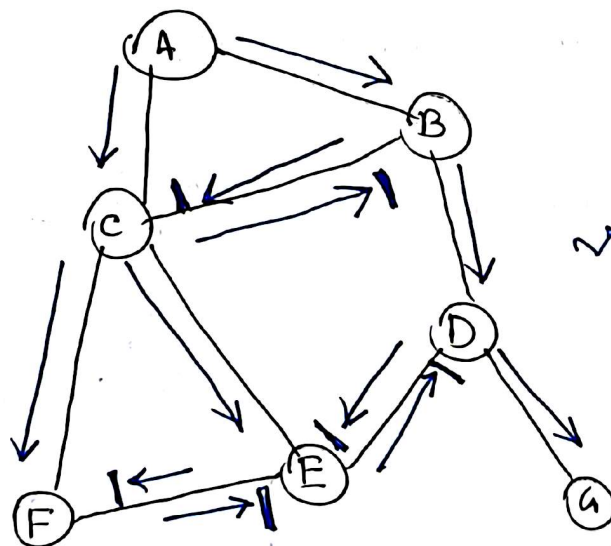
Discard duplicated packets

If same packet comes from multiple neighbours, what to choose?

- Packets have roots
- Source is only one

Try to follow from packet to source & find minimum root

↳ হুবহু minimum root খুঁজে accept করবে & neighbourকে broadcast করে দিবে।



#WiMax
Self Study
or
extended class

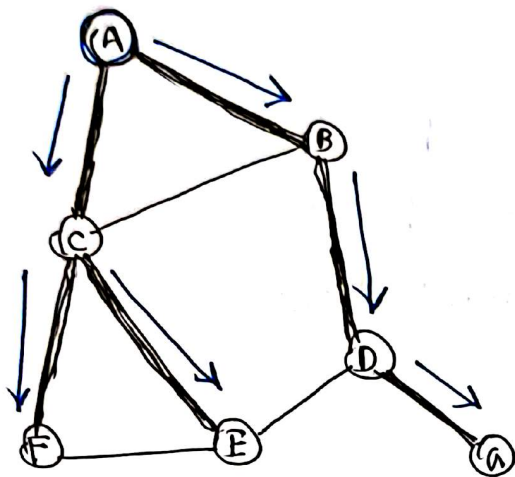
→ pkt will be forwarded.

→ ❌ pkt will not be forwarded beyond receiving router.

4.44: Reverse Path Forwarding

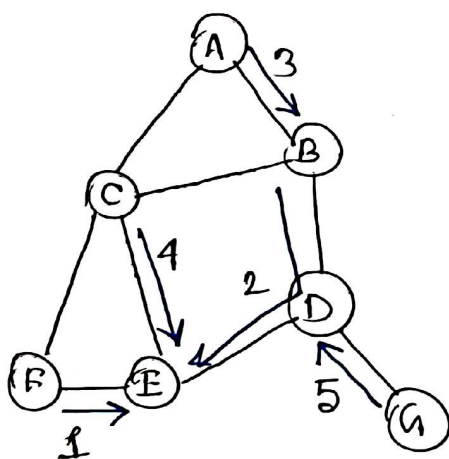
Spanning Tree Broadcast

mShayan

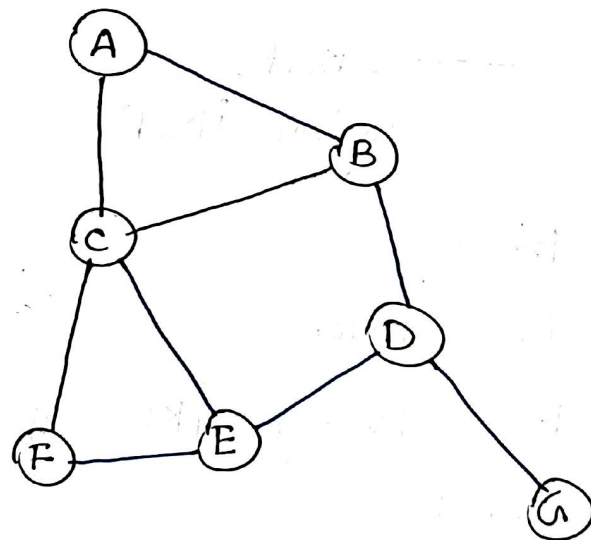


Broadcast Initiated at A.

Algorithm:
Center-Based Algorithm
to find minimum
spanning tree

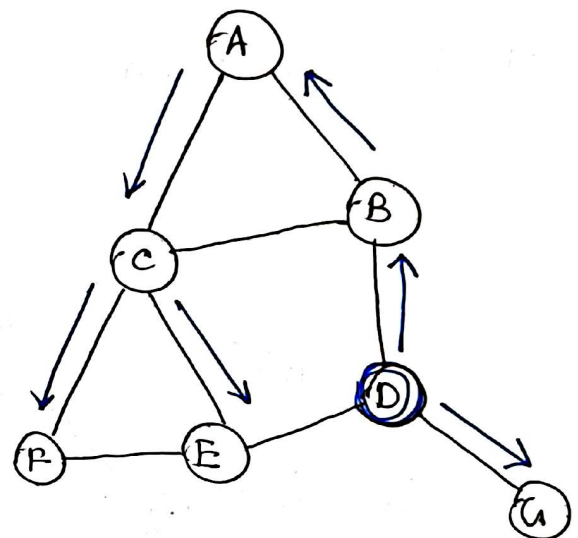


(a). Step-wise
Construction of
Spanning Tree



(b). Constructed Spanning
tree

Graph & Tree
Vertices & Edges
Spanning Tree
Minimum Spanning Tree



Broadcast initiated
at D.

#Border Gateway Protocol (BGP)

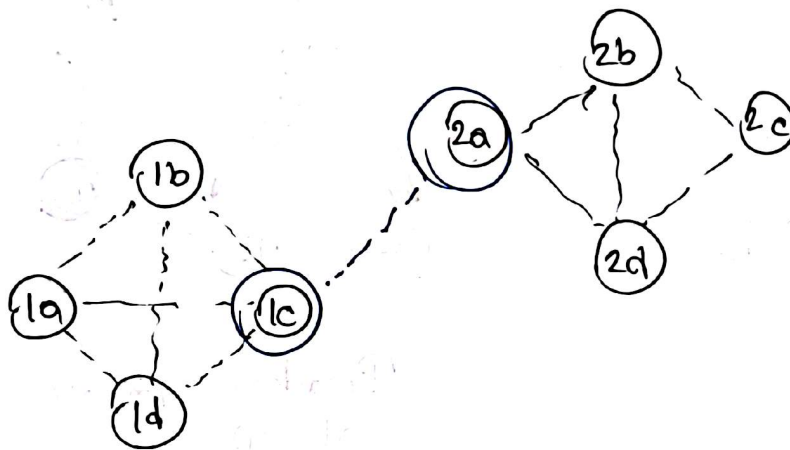
AS → Autonomous System

Intra - Autonomous Routing Protocol

Inter - Autonomous Routing Protocol

✓ VLSM (Variable Length Subnet Mask)

eBGP, iBGP connection



--- eBGP

--- logical iBGP

BGP Basics

└ BGP Session

BGP Protocol Messages [RFC 4371]

- ① OPEN
- ② UPDATE
- ③ KEEPALIVE
- ④ NOTIFICATION

#Path Attributes of BGP Routes

Shayan

BGP advertised route: prefix + attributes

prefix: destination being advertised

two important attributes:

AS-PATH: list of ASes through which prefix advertisement has passed.

NEXT-HOP: indicates specific internal AS router to next-hop AS.

Policy Based Routing: [x]